

4.2.4 Bulk and surface properties of matter including nanoparticles (chemistry only)

4.2.4.1 Sizes of particles and their properties

Content	Key opportunities for skills development
Nanoscience refers to structures that are 1–100 nm in size, of the order of a few hundred atoms. Nanoparticles, are smaller than fine particles ($\text{PM}_{2.5}$), which have diameters between 100 and 2500 nm (1×10^{-7} m and 2.5×10^{-6} m). Coarse particles (PM_{10}) have diameters between 1×10^{-5} m and 2.5×10^{-6} m. Coarse particles are often referred to as dust. As the side of cube decreases by a factor of 10 the surface area to volume ratio increases by a factor of 10. Nanoparticles may have properties different from those for the same materials in bulk because of their high surface area to volume ratio. It may also mean that smaller quantities are needed to be effective than for materials with normal particle sizes.	WS 1.2, 1.4, 4.1, 4.2, 4.3 4.4, 4.5 MS 2h Make order of magnitude calculations. MS 5c Calculate areas of triangles and rectangles, surface areas and volumes of cubes.

Content	Key opportunities for skills development
Students should be able to compare 'nano' dimensions to typical dimensions of atoms and molecules.	MS 1b Recognise and use expressions in standard form. MS 1c Use ratios, fractions and percentages. MS 1d Make estimates of the results of simple calculations.

4.2.4.2 Uses of nanoparticles

Content	Key opportunities for skills development
Nanoparticles have many applications in medicine, in electronics, in cosmetics and sun creams, as deodorants, and as catalysts. New applications for nanoparticulate materials are an important area of research. Students should consider advantages and disadvantages of the applications of these nanoparticulate materials, but do not need to know specific examples or properties other than those specified.	
Students should be able to: • given appropriate information, evaluate the use of nanoparticles for a specified purpose • explain that there are possible risks associated with the use of nanoparticles.	WS 1.3, 1.4, 1.5